



CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)

CHARUSAT Campus, Changa – 388 421.

M Sc (BIOTECHNOLOGY)

SEMESTER III EXAMINATION (BACKLOG)

BT803: Environmental Biotechnology and Bioinformatics

Date: 23/4/2011

Time: 10.00 AM to 1.00 PM

Total Marks: 70

Instructions to the candidates:

Provide all necessary information as required in the answer sheet.

1. Attempt all questions.
2. Questions in **Part A** should be answered in the question paper itself. Please **do not write** your candidate ID number on the question paper for Part A.
3. Write answers for Section I and Section II of **Part B** in separate answer sheets.
4. Use of cell phones is strictly prohibited.
5. Use of non-programmable calculators may be allowed if needed.

PART A

Total Marks: 20

1. Pumped systems, grab samples and preconcentration are the type of sampling methods used for Monitoring (a) water (b) air (c) soil (d) a, b and c
2. Metals can be detected by using (a) Mass Spectrophotometry (b) HPLC (c) GC (d) Atomic adsorption spectrophotometry
3. Which of the following is the nonprobability based sampling method?
(a) Random (b) systematic (c) stratified (d) judgement
4. EC_{50} is the concentration of the pollutant that reduces growth by 50%. [TRUE/FALSE]
5. When environmental reactivity of a chemical is reduced by its complexation with other materials, the process is known as
(a) Sequestration (b) Biodegradation (c) Bioremediation (d) Bioavailability
6. Enhanced biodegradation of organic pollutant upon addition of a second carbon source is known as cometabolism. [TRUE/FALSE]
7. Lignin is the polymer of the following monomers
(a) p-coumaric alcohol (b) coniferyl alcohol (c) sinapyl alcohols (d) all the above
8. Monooxygenases incorporate one atom of dioxygen (O_2) into the substrate. [TRUE/FALSE]
9. The sludge that settles well initially but starts to rise again in the settling tank is known as
(a) pin point (b) bulking (c) anoxic (d) all of the above
10. Shredding and pulverization of solid waste increases the rate of biodegradation by increasing the following of the pollutants:
(a) surface area available (b) bioavailability (c) bioaccumulation (d) all of the above
11. The following is **NOT** the part of activated sludge process:
(a) primary treatment (b) secondary settling tank (c) aeration tank (d) none of the above

12. The following process involves formation of biofilms for the degradation of pollutants present in industrial effluents
(a) Activated sludge (b) trickling filters (c) land farming (d) all of the above
13. First protein sequence data bank is known as
(a) Atlas of Protein Sequence and Structure (b) Protein sequence databank (c) Protein databank (d) Amino acid and protein sequence databank
14. SNP database is an example of
(a) Primary database (b) Secondary database (c) Tertiary database (d) Specialized database
15. _____ provides integrated access to nucleotide and protein sequence database
(a) Entrez---correct (b) BLAST (c) pBLAST (d) nBLAT
16. OMIM stands for
(a) Online Mendelian Inheritance in Man (b) Online Mendelian Inheritance in Mammal (c) Online Mendelian Inheritance in Mouse (d) All the above
17. Following serves as molecular fossils in phylogenetics
(a) Cholesterol (b) Triglyceride (c) Glucose (d) DNA
18. Bifurcation of branches in a phylogenetic tree indicates
(a) Polytoomy (b) Multitomy (c) Dichotomy--- correct (d) Bichotomy
19. A good protein structures should have
(a) low R factor (b) high R factor (c) R factor as zero (d) negative R factor
20. ClustalW is a
(a) Sequence Alignment Tool (b) Sequence submission tool (c) Both (d) None



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PART B

Total Marks: 50

Section I

Q 1 Answer any two [10]

- A Note on Sampling
- B Global warming
- C Note on Biosensor
- D Acid Rain

Q 2 Write on the following [4]

- A Depict the pathway for aromatic hydrocarbon degradation pathway and briefly explain it.

OR

Role of laccase and LiP in remediation of pollutant.

- B Write brief note on following (**any of two**) [6]
 - (i) Cometabolism
 - (ii) Xenobiotic compounds
 - (iii) Anaerobic degradation
 - (iv) Monooxygenase enzymes

Section II

Q 3 A Write note on following (**any two**) [6]

- (i) Flocculation of microorganisms in activated sludge.
- (ii) Bioremediation by landfarming
- (iii) Phytoremediation
- (iv) Biosparging

- B Explain briefly (**any two**) [4]
 (i) The biology of ASP
 (ii) Sloughing in Trickling filter
 (iii) COD and bCOD
- Q 4 A What is a GenBank flat file? [3]
 OR
 Write a short note on OMIM.
- B What is a database? Describe in brief various types of biological databases. [3]
 OR
 What are the different bioinformatics tools available at EBI. Describe in brief.
- C Write a short note on [2]
 TPA OR Gene annotation
- D Describe in brief various tools of data mining in bioinformatics world. [2]
 OR
 Tools for sequence analysis
- Q 5 A Answer Any two [4]
 (i) What is the significance of studying Phylogenetics?
 (ii) What is bootstrapping?
 (iii) Mention various steps involved in Phylogenetic Tree Construction.
 (iv) What is an Intermolecular algorithm for protein structure comparison?
- B Answer Any two [6]
 (i) What characteristics a good protein structures should have to obtain high quality 3D protein image?
 (ii) Write a short note "SCOP"
 (iii) Describe in brief the two Protein structure Classification Schemes.
 (iv) Describe in brief six softwares that aid in visualizing 3D protein structure.